Design of MDP Convolutional Codes and Maximally Recoverable Codes Through the Lens of Matrix Completion

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Abstract

The matrix completion problem provides a unifying lens through which many fundamental problems in coding theory can be viewed. In this paper, we investigate Locally Recoverable Codes (LRCs) with Maximal Recoverability (MR) and Maximum Distance Profile (MDP) convolutional codes in the framework of matrix completion. In particular, we present techniques that are general enough to provide constructions for both types of codes. A common feature of our code constructions is the sparsity of their generator matrices and the property that a large number of the entries of the generator matrices are elements of a small subfield of a larger extension field.

Key Words : Matrix Completion, Convolutional codes, Maximally recoverable codes.

1 Introduction

The *matrix completion problem* provides a powerful and unifying perspective for understanding a broad class of fundamental problems in coding theory and related areas. In particular, many code design problems, including the construction of convolutional codes, distributed storage codes, and codes for

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wireless communication, can be naturally formulated as variants of the matrix completion problem. In its most general formulation, the problem considers a partially specified matrix subject to structural constraints, such as column dependencies, and block-structure requirements, together with unconstrained (free) entries. These constraints can be captured algebraically by specifying a set of polynomial equations over the matrix entries and fixing selected entries to given values. The goal of the matrix completion problem is to assign values to the free entries to satisfy specified algebraic objectives, such as ensuring the non-vanishing of specified full-size minors, while minimizing the required finite field size. The overarching challenge is to develop principled matrix completion frameworks that apply uniformly across a wide range of constraint patterns. One class of matrix completion problems concerns the construction of a generator matrix of an MDS code that has a prescribed zero pattern (support constraint). The related GM-MDS conjecture, introduced in [DSY14] was resolved in full in [Lov21, YH19].

Also the design of *Maximum Distance Profile* (MDP) convolutional codes (commonly referred to as MDP codes) can be viewed as a variation of the matrix completion problem [DLM^{+}25]. Convolutional codes are important for low-delay encoding and decoding of a stream of data. MDP codes are a class of convolutional codes that allows for best possible error-correction in this setting [TRS12]. The sliding generator matrix of a convolutional code takes a specific form in which multiple copies of blocks are arranged in a repeating sliding pattern and some entries are fixed to zero. While this structural constraint does not allow every full-size minor to be non-zero, sliding generator matrices of MDP codes have the property that each full size minor that can be nonzero, given the prescribed structural requirement, is nonzero, [GLRS06]. There exist several constructions for MDP codes, however mostly requiring large finite fields; see e.g., [ANNR24, Che23, LCEL23, ANP13].

Another problem that can be viewed through the lens of Matrix Completion is the design of *Locally Recoverable Codes* (LRCs) that are *Maximally Recoverable* (MR). The local recoverability property implies the existence of a collection of *repair groups* such that each group enables local recovery of its symbols; that is, symbol(s) in a repair group can be recovered using only the symbols within that group. MR-LRCs provide recoverability from the largest possible set of erasure patterns (see [BHH13, GHJY14, GG22, CMST22, MP22] and references therein) among linear block codes fulfilling certain locality constraints. In this paper, we focus on a class of MR LRCs with unequal locality, i.e., codes in which the subcodes that correspond to different repair groups might have different sizes and dimensions, in addition to a set of global parity symbols which are not included in the repair groups.

We present a matrix-completion technique that is broad enough to encompass the construction of both code families. A unifying aspect of these constructions is a sparse generator matrix with elegant structural properties including that certain submatrices of the generator matrix are constructed using a small subfield of a larger extension field. Such a structure facilitates efficient encoding and decoding procedures. In Section 2, we provide preliminaries on matrix completion, MR-LRCs and MDP codes. In Section 3, we present a construction for MR-LRCs with unequal localities for certain parameters. In Section 4, we present constructions for MDP codes with certain parameters. In Section 5, we explain commonalities of these constructions and mention some related matrix completion problems to be considered for future research.

2 Preliminaries

We begin with a general matrix completion problem and then consider specific instances, i.e., the construction of certain MR-LRCs and MDP codes.

Definition 1. *Let M be a $K \times N$ matrix whose entries $m_{i,j}$ for $i = 1, \ldots, K$, $j = 1, \ldots, N$ are considered as variables. Define $\mathbb{F}_q[M] := \mathbb{F}_q[m_{1,1}, \ldots, m_{K,N}]$ and consider a set of polynomials $f_\ell(M) \in \mathbb{F}_q[M]$ for $\ell = 1, \ldots, s$. We say that a **solution** to the **matrix completion problem** corresponding to $f_1, \ldots, f_s$ is a set of values for $m_{i,j} \in \mathbb{F}_q$ satisfying the following conditions:*

1. *$f_\ell(M) = 0$ for $\ell = 1, \ldots, s$ (these equations are called **constraints** of the matrix completion problem);*
2. *all **non-trivial full-size minors**,* i.e., *the $K \times K$ minors of M that are not nilpotent in the ring $\mathbb{F}_q[M]/\langle f_1(M), \ldots, f_s(M)\rangle$, are non-zero at the values chosen for $m_{i,j}$ (f is nilpotent if there exists a positive integer n such that $f^n = 0$).*

2.1 Maximally Recoverable Locally Recoverable Codes with Unequal Locality (MR-LRCs with UL)

In this paper, we focus on Locally Recoverable Codes with Unequal Locality (LRCs with UL) [KS16, ZY16], defined as follows.

Definition 2. *Let $\mathcal{C}$ be a non-degenerate linear block code of length L. We say that a codeword symbol c_j for $j \in \{1, \ldots, L\}$ has **locality** k_j if there exists a **repair group** $R_j \subset \{1, \ldots, L\}$ such that $j \in R_j$ and $\mathcal{C}|_{R_j}$ forms a $[|R_j|, k_j, \delta_j]$-code; if $\delta_j > 1$ for all j, then we call it a **Locally Recoverable Code with Unequal Locality**.*

There exists a modified Singleton bound on the minimum distance of LRCs with UL [KS16, ZY16]. We extend LRCs with UL to MR-LRCs with UL by requiring maximum recoverability, i.e., all erasure patterns that can be recovered under the given locality constraints should be recovered. This property is considerably stronger than reaching the modified Singleton bound and captures the code's full erasure-correction potential.

Definition 3. *For $\ell, h, n_i, k_i \in \mathbb{N}$ with $n_i \geq k_i$ for $i = 0, \ldots, \ell - 1$, let $\mathcal{C} \subseteq \mathbb{F}_q^N$ be an $\mathbb{F}_q$-linear code of length $N = n_0 + \cdots + n_{\ell-1} + h$ and dimension $K = k_0 + \cdots + k_{\ell-1}$. Set $\mathcal{G}_i := \{1 + \sum_{t=0}^{i-1} n_t, \ldots, \sum_{t=0}^{i} n_t\}$ for $i = 0, \ldots, \ell-1$. Note that these sets form a partition of $\{1, \ldots, N-h\}$ and $|\mathcal{G}_i| = n_i$. We say that $\mathcal{C}$ is **MR-LRC with UL** if*

1. *$\dim(\mathcal{C}|_{\mathcal{G}_i}) \leq k_i$, i.e., each codeword symbol indexed by $\mathcal{G}_i$ has locality k_i;*
2. *for each **admissible** puncturing S, i.e., for each puncturing where exactly $n_i - k_i$ coordinates in $\mathcal{G}_i$ are deleted, the punctured code $\mathcal{C}|_S$ is an $[h + K, K]$-MDS code.*

*The sets $\mathcal{G}_i$ are called **local repair groups** and the last h coordinates of a codeword are called the **global parities**.*

Remark 1. *Note that Definition 3 implies that $\mathcal{C}|_{\mathcal{G}_i}$ is an $[n_i, k_i]$-MDS code for $i = 0, \ldots, \ell - 1$. Also note that in contrast to the MR-LRCs considered in the literature before (see e.g. [GG22]), by considering MR-LRCs with UL we allow the local repair groups to have different sizes n_i and localities k_i (while the h global parities have locality K) but require that $k_0 + \cdots + k_{\ell-1}$ equals the dimension K of the full code. MR-LRCs are also known as partial MDS codes [BHH13]. In [HTN20], partial MDS codes with different sizes n_i are considered but still it is required that all localities k_i are equal.*

Remark 2. *The problem of constructing MR-LRCs with UL can be seen as matrix completion problem where Condition 1 of Definition 3 provides the constraints and Condition 2 describes the non-trivial full-size minors of a generator matrix M of $\mathcal{C}$. More precisely, if M_i is the $K \times n_i$ submatrix of M given by the columns indexed by $\mathcal{G}_i$, Condition 1 of Definition 3 is equivalent to the constraints $f_{i,S,T}(M) = 0$ for all $S \subset \{1, \ldots, K\}$, $T \subset \{1, \ldots, n_i\}$ with $|S| = |T| = k_i + 1$, for $i = 0, \ldots, \ell - 1$, where $f_{i,S,T}(M)$ is the minor of M_i corresponding to the rows and columns indexed by S and T respectively.*

2.2 Convolutional Codes

Definition 4. *For $k \leq n$, an (n, k) **convolutional code** $\mathcal{C}$ is defined as $\mathbb{F}_q[z]$-submodule of $\mathbb{F}_q[z]^n$ of rank k. A polynomial matrix $G(z) \in \mathbb{F}_q[z]^{k \times n}$ of full (row) rank such that $\mathcal{C} = \{v(z) = u(z)G(z) \in \mathbb{F}_q[z]^n \mid u(z) \in \mathbb{F}_q[z]^k\}$ is called **generator matrix** of $\mathcal{C}$. The maximum (polynomial) degree of the full-size minors of a generator matrix is called the **degree** δ of $\mathcal{C}$. An (n, k) convolutional code of degree δ is denoted as (n, k, δ) convolutional code.*

In contrast to linear block codes not all convolutional codes admit a representation via a parity-check but only those with a generator matrix $G(z) \in \mathbb{F}_q[z]^{k \times n}$ that is left-prime, i.e., $G(z)$ is full rank for all $z \in \overline{\mathbb{F}}_q$, where

$\overline{\mathbb{F}}_q$ denotes the algebraic closure of $\mathbb{F}_q$; see [Yor97]. Such convolutional codes are called **non-catastrophic**.

The structure of convolutional codes makes them very suitable for low-delay applications and the crucial distance measure for such applications are the column distances.

Definition 5. *For $j \in \mathbb{N}_0$, the* ***j-th column distance*** *of a convolutional code $\mathcal{C}$ is $d_j^c(\mathcal{C}) := \min\left\{\sum_{t=0}^{j} wt(v_t) \mid v(z) = \sum_{i\in\mathbb{N}_0} v_i z^i \in \mathcal{C} \text{ with } v_0 \neq 0\right\}$.*

For the j-th column distance, one has the following upper bound.

Theorem 1. *[GLRS06] Let $\mathcal{C}$ be an (n,k,δ) convolutional code. Then, for $j \in \mathbb{N}_0$, $d_j^c(\mathcal{C}) \leq (n-k)(j+1)+1$. If $d_j^c = (n-k)(j+1)+1$ for some $j \in \mathbb{N}_0$, then $d_i^c = (n-k)(i+1)+1$, for $i \leq j$. Moreover, $d_j^c(\mathcal{C}) = (n-k)(j+1)+1$ implies $j \leq L := \left\lfloor \frac{\delta}{k} \right\rfloor + \left\lfloor \frac{\delta}{n-k} \right\rfloor$.*

Definition 6. *An (n,k,δ) convolutional code $\mathcal{C}$ is called* ***maximum distance profile (MDP)*** *if $d_L^c(\mathcal{C}) = (n-k)(L+1)+1$.*

Let $G(z) = \sum_{i=0}^{\mu} G_i z^i$ with $G_\mu \neq 0$, where $\mu := \deg(G(z))$ is called **memory** of $G(z)$. A convolutional code that possesses a generator matrix $G(z)$ with $\mu = 1$ is called **unit-memory** if $rk(G_1) = k$ and **partial unit-memory** if $rk(G_1) < k$. The **sliding generator matrix** is defined as

$$G_j^c := \begin{bmatrix} G_0 & \cdots & G_j \\ & \ddots & \vdots \\ \mathbf{0} & & G_0 \end{bmatrix} \in \mathbb{F}_q^{(j+1)k\times(j+1)n} \quad \text{for} \quad j \in \mathbb{N}_0,$$

with $G_i = 0$ for $i > \mu$. Next, we present a criterion to check whether a convolutional code is MDP.

Theorem 2. *[GLRS06] Let $\mathcal{C}$ be a convolutional code with generator matrix $G(z) = \sum_{i=0}^{\mu} G_i z^i \in \mathbb{F}_q[z]^{k\times n}$ and $0 \leq j \leq L$. The following statements are equivalent:*

(a) $d_j^c(\mathcal{C}) = (n-k)(j+1)+1$.

(b) All non-trivial full-size minors of $G_j^c \in \mathbb{F}_q^{(j+1)k\times(j+1)n}$ are nonzero, i.e., all minors formed by columns with indices $1 \leq t_1 < \cdots < t_{(j+1)k}$, where $t_{sk+1} > sn$, for $s = 1, 2, \ldots, j$.

Note that the condition of the previous theorem implies that G_0 has to be an MDS matrix. In this paper, we only consider partial unit-memory convolutional codes with $k > n-k = \delta$, i.e., $L = 1$; note that this class of codes is also considered in [LCEL23, Che23].

Remark 3. *The construction of MDP codes can be stated as a matrix completion problem for $M = G_L^c$. For $L = 1$, the constraints are given by $m_{i,j} = 0$ for $(i,j) \in \{k+1,\ldots,n\} \times \{1,\ldots,n\}$ and $m_{i,j} - m_{i+k,j+n} = 0$ for $(i,j) \in \{1,\ldots,k\} \times \{1,\ldots,n\}$.*

3 Construction of MR-LRCs with UL

In the following, we present a construction for MR-LRCs with UL with very sparse generator matrices. To this end, for some finite field $\mathbb{F}$, we consider a generator matrix of the form:

$$G := \begin{bmatrix} G_0 & 0 & 0 & \dots & 0 & P_0 \\ 0 & G_1 & 0 & \dots & 0 & P_1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & G_{\ell-1} & P_{\ell-1} \end{bmatrix} \tag{1}$$

with $G_i \in \mathbb{F}^{k_i \times n_i}$ and $P_i \in \mathbb{F}^{k_i \times h}$ for $i = 0, \dots, \ell-1$. Moreover, if a message $m \in \mathbb{F}^K$ is partitioned as $m = (m_0, \dots, m_{\ell-1})$ with **information blocks** $m_i \in \mathbb{F}^{k_i}$ for $i = 0, \dots, \ell-1$, then MR-LRCs with UL of this form have the property that m_i can be recovered from $m_i G_i$ if not more than $n_i - k_i$ erasures happened in the coordinates with indices in $\mathcal{G}_i$. This property generalizes the notion of information symbol locality (see e.g. [PKLK12]) to what we call **information block locality**. In this way we further specify the matrix completion problem given in Remark 2 via considering generator matrices of the form (1) where we additionally fix G_i to be Cauchy matrices. In this setup, solving the corresponding matrix completion problem asks, given the structure of G and Cauchy matrices G_i, to find P_i such that all non-trivial full-size minors of G are nonzero.

Theorem 3. *Let $h \leq \min\{k_i,\ 0 \leq i \leq \ell - 1\}$ and let $G_i \in \mathbb{F}_q^{k_i \times n_i}$, for $i = 0, \dots, \ell-1$, be Cauchy matrices and*

$$P_i := \begin{bmatrix} \mathrm{Diag}(\alpha^0, \alpha^i, \alpha^{2i}, \dots, \alpha^{(h-1)i}) \\ \mathbf{0}_{(k_i-h)\times h} \end{bmatrix} \tag{2}$$

for some primitive element $\alpha \in \mathbb{F}_{q^d}$. If $h = 1$ and $d \geq 1$, or if $h > 1$ and

$$d \geq (\ell - 1) \cdot \left\lfloor \frac{h}{2} \right\rfloor \cdot \left\lceil \frac{h}{2} \right\rceil + 1 =: D,$$

then the corresponding code is MR-LRC with UL over $\mathbb{F}_{q^d}$.

Proof. We provide an outline of the proof, for more details see [DLSS26]. Recall that we need to show that if we puncture the i-th local repair group $\mathcal{G}_i$ in $n_i - k_i$ positions, then the punctured code is MDS. Therefore, we can assume that G_i's are square Cauchy matrices and need to show that the corresponding code is MDS. All fullsize minors of G are polynomials over $\mathbb{F}_q$ in α and we will show that all non-trivial full-size minors are nonzero polynomials whose difference between the degrees of the maximal and minimal nonzero term is at most $D - 1$.

Consider first the case that there are no erasures in the last h columns of G, i.e., no erasures in the global parity symbols.

For $i = 0, \dots, \ell-1$, denote by x_i the number of erasures in the columns corresponding to G_i (after the puncturing that causes that the G_i are square matrices), i.e., $x_0 + \cdots + x_{\ell-1} = h$. When calculating the corresponding minor, each nonzero term in the determinant corresponds to choosing x_i entries in P_i, for all $i = 0, \dots, \ell-1$, where all the choices (over all P_i) are from distinct columns (out of the last h columns) of G. Let $w_{i,j}$ be the j^{th} choice for P_i and let $T_{\vec{x}}$ be set of all such choices. Then, the minor has the form

$$\sum_{w \in T_{\vec{x}}} M_w \prod_{i=0}^{\ell-1} \alpha^{iw_{i,0}} \cdot \alpha^{iw_{i,1}} \cdot \ \cdots \ \cdot \alpha^{iw_{i,x_i-1}} = \sum_{w \in T_{\vec{x}}} M_w \alpha^{\sum_{i=0}^{\ell-1} i \sum_{j \in [x_i]} w_{i,j}} \tag{3}$$

where M_w is a product of minors of Cauchy matrices and hence, nonzero. We obtain the minimal exponent in α when choosing $w_{\ell-1,j} = j, w_{\ell-2,j} = x_{\ell-1} + j, \dots, w_{0,j} = x_1 + x_2 + \cdots + x_{\ell-1} + j$. On the other hand, we obtain the maximal exponent in α for $w_{0,j} = j, w_{1,j} = x_0 + j, \dots, w_{\ell-1,j} = x_0 + x_1 + \cdots + x_{\ell-2} + j$. Therefore, the difference between the degrees of the maximal and minimal nonzero term is equal to

$$\sum_{i=0}^{\ell-1} i \sum_{j \in [x_i]} w_{i,j}^{\max} - w_{i,j}^{\min} = \sum_{0 \le i < j \le \ell-1} (j-i) x_i x_j =: Q_h(x_0, \dots, x_{\ell-1})$$

Hence, we have to maximize the quadratic form $Q_h(x_0, \dots, x_{\ell-1})$ subject to the constraints $x_0 + \cdots + x_{\ell-1} = h$ and $x_0, \dots, x_{\ell-1} \in \mathbb{N}_0$. One can show that Q_h attains this maximum for $x_0 = \left\lfloor \frac{h}{2} \right\rfloor$, $x_1 = \cdots = x_{\ell-2} = 0$, $x_{\ell-1} = \left\lceil \frac{h}{2} \right\rceil$ and this maximum is equal to $D = (\ell-1) \cdot \left\lfloor \frac{h}{2} \right\rfloor \cdot \left\lceil \frac{h}{2} \right\rceil$.

It is easy to see that the difference between maximal and minimal degree in α of the nonzero terms of any non-trivial full-size minor of G is smaller than $D-1$ if we have erasures in the last h columns of G. □

Our construction has the advantage of a very sparse generator matrix leading to low encoding complexity and good update efficiency as explained in the following. To analyze the encoding complexity for the codes of the previous theorem, we follow the standard convention to only count the number of multiplications, but in our case, the number of additions is comparable. To calculate mG for $m \in \mathbb{F}_{q^d}^K$ and G as in (1), for each $i = 0, \dots, \ell-1$, we need to multiply a vector from $\mathbb{F}_{q^d}^{k_i}$ with an $k_i \times n_i$ Cauchy matrix with entries in $\mathbb{F}_q$, which requires $O\left(\sum_{i=0}^{\ell-1} n_i \log^2 n_i\right)$ multiplications in $\mathbb{F}_{q^d}$ (see [GGS87]), and additionally we need to multiply $k_0 + \cdots + k_{\ell-1}$ elements of the form α^t for some $t \in \mathbb{N}_0$ with an element of $\mathbb{F}_{q^d}$. Hence, we get the following result:

Theorem 4. *The encoding complexity for our construction given in Theorem 3 is* $O\left(\sum_{i=0}^{\ell-1} n_i \log^2 n_i\right)$.

Moreover, if some message symbol needs to be updated, i.e., its value to be changed, then not all of the codeword symbols need to be recomputed; one only has to update the corresponding local group and exactly one global parity symbol, since the diagonal structure of the matrices $P_0, \dots, P_{\ell-1}$ implies that each message symbol is only involved in the calculation of one global parity symbol.

4 Construction of partial unit-memory MDP codes

Lemma 5. *Let* $G(z) = G_0 + G_1 z \in \mathbb{F}_q[z]^{k\times n}$*, where* G_0 *is an MDS matrix and* G_1 *is of the form* $G_1 = [X\ \mathbf{0}_{k\times k}]$*. Then, the corresponding convolutional code is non-catastrophic.*

Proof. For all $\hat{z} \in \overline{\mathbb{F}}_q$, the last k columns of $G(\hat{z})$ are equal to the last k columns of G_0 and hence, $G(\hat{z})$ is full rank for all $\hat{z} \in \overline{\mathbb{F}}_q$. □

In the construction for partial unit-memory MDP codes from [DLM+25],

$$X := \begin{bmatrix} \mathrm{Diag}(\alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-k}) \\ \mathbf{0}_{(2k-n)\times(n-k)} \end{bmatrix} \in \mathbb{F}_{q^d}^{k\times(n-k)}, \tag{4}$$

where $\alpha \in \mathbb{F}_{q^d}$ has minimal polynomial over $\mathbb{F}_q$ of degree d.

Theorem 6. *[DLM+25] Let* $k > n-k = \delta$ *and* $d = \left\lceil \frac{\delta^2-1}{4} \right\rceil + 1$*. Let* $G_0 \in \mathbb{F}_q^{k\times n}$ *be a superregular matrix and* $G_1 = [X\ \mathbf{0}_{k\times k}] \in \mathbb{F}_{q^d}^{k\times n}$ *with* X *given by equation* (4)*. Then, all nontrivial minors of* G_1^c *are nonzero, i.e., we get an* (n,k,δ)*-MDP code over* $\mathbb{F}_{q^d}$*.*

The previous theorem can be seen as special case of the matrix completion problem from Remark 3 in the sense that we additionally fix G_0 to be **superregular** (i.e., a matrix whose every minor is nonzero) and G_1 to be of the form $[X\ \mathbf{0}_{k\times k}]$ for some matrix X to be determined. In the following, we present alternative constructions for partial unit-memory MDP codes using a Vandermonde matrix for G_0 and a different structure for the matrix X. These constructions have the advantage that they require smaller finite fields but the disadvantage that we can only cover the special cases $\delta = 2$ and $\delta = 3$. In particular, the field size for our previous construction, [DLM+25], is $q = O((n+k)^\delta)$, whereas the following construction has $q = O(n^\delta)$. For $\beta_1, \dots, \beta_{n-k} \in \mathbb{F}_{q^{n-k}}$, we define

$$\hat{X} := \begin{bmatrix} \mathbf{0}_{(2k-n)\times(n-k)} \\ T \end{bmatrix} \quad \text{with} \quad T := \begin{bmatrix} \beta_1 & 0 & \cdots & 0 \\ \beta_2 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \beta_{n-k} & \cdots & \beta_2 & \beta_1 \end{bmatrix}. \tag{5}$$

A proof for the following theorem can be found in [DLSS26].

Theorem 7. *Let $q \geq n \geq 4$, $k = n-2$. Let $G_0 \in \mathbb{F}_q^{k\times n}$ be a Vandermonde matrix and $\hat{X}$ a $k \times 2$ matrix as in (5) where $\beta_1, \beta_2 \in \mathbb{F}_{q^2}$ are linearly independent over $\mathbb{F}_q$. Then, all nontrivial minors of G_1^c are nonzero, i.e., we get an $(n,k,2)$ MDP code over $\mathbb{F}_{q^2}$.*

Theorem 8. *Let $q \geq n$, $k = n-3$ and $G_0 \in \mathbb{F}_q^{k\times n}$ be a Vandermonde matrix. Let $G_1 = [\hat{X}\ \mathbf{0}_{k\times k}]$ with $\hat{X}$ as in (5) with $\beta_1 = z$, $\beta_2 = z^2$, $\beta_3 = 1$ where $z \in \mathbb{F}_{q^3} \setminus \mathbb{F}_q$ such that the minimal polynomial of z has constant coefficient unequal to -1. Then, all nontrivial minors of G_1^c are nonzero, i.e., we get an $(n,k,3)$ MDP code over $\mathbb{F}_{q^3}$.*

Proof. We provide only an outline of the proof; full details are available in [DLSS26]. For integers i, j, ℓ, we consider any $2k \times 2k$ submatrix M of G_1^c by choosing i columns out of first n columns, j columns out of next $n-k$ and ℓ columns out of last k columns. If M corresponds to a non-trivial minor, then $i + j + \ell = 2k$ where $1 \leq i \leq k,\ 0 \leq j \leq n-k = 3$. Thus,

$$M = \begin{bmatrix} U & V \\ \mathbf{0} & W \end{bmatrix} \quad \text{with} \quad V = [\hat{V}\ \mathbf{0}_{k\times \ell}],$$

where U is a submatrix of G_0 of size $k \times i$, V is a submatrix of G_1 of size $k \times (j+\ell)$, W is a submatrix of G_0 of size $k \times (j+\ell)$. To show $\det(M) \neq 0$, we consider different cases, for which we use the abbreviation (i,j,ℓ).

Case $(k, j, k-j)$: This case is clear from the fact that G_0 is an MDS matrix.

Case $(k-j, j, k)$: In this case, $\det(M) = \det([U\ \hat{V}]) \cdot \det(W)$ and it suffices to show that $\det([U\ \hat{V}]) \neq 0$, i.e., to show $([G_0\ \hat{X}])$ is an MDS matrix. We consider 3 cases for $j = 1,2,3$ respectively and show that $\det([U\ \hat{V}]) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$ with $a_i \in \mathbb{F}_q$ such that $a_i \neq 0$ for (at least one) $i \in \{0,1,2\}$ and $a_3 = 0$ or $\det([U\ \hat{V}]) \neq f(z)$, the minimal polynomial of z. In the remaining cases, we write $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ where A, B, C, D are $k \times k$ matrices and use the fact that $\det(M) \neq 0 \iff \det(A - BD^{-1}C) \neq 0$.

Case $(k-1, 2, k-1)$: In this case, $A = [\tilde{g}_1\ \cdots\ \tilde{g}_{k-1}\ x_{i_1}]$, $B = [x_{i_2}\ 0\ \cdots\ 0]$, $C = [0\ \cdots\ 0\ g_{i_1}]$ and $D = [g_{i_2}\ g_{i_3}\ \cdots\ g_{i_{n-2}}]$ where $\tilde{g}_i$ are different columns from G_0, x_{i_1}, x_{i_2} correspond to 2 columns of X, and g_{i_j} is the i_j-th column of G_0 with $1 \leq i_1 < i_2 < \cdots < i_{n-3} < i_{n-2} \leq n$. For different choices of $i_1 < i_2$ from {1,2,3}, we observe that $\det(A - BD^{-1}C) = az^2 + bz + c$ with either $a \in \mathbb{F}_q^*$ or $c \in \mathbb{F}_q^*$. Hence, $\det(M) \neq 0$.

Case $(k-1, 3, k-2)$: Here, $A = [\tilde{g}_1\ \cdots\ \tilde{g}_{k-1}\ x_1]$, $B = [x_2\ x_3\ 0\ \cdots\ 0]$, $C = [0\ \cdots\ 0\ g_1]$ and $D = [g_2\ \ g_3\ \ g_{i_4}\ \cdots\ g_{i_{n-2}}]$ where $\tilde{g}_i$ are different columns from G_0 and g_{i_j} the i_j-th column of G_0, $4 \leq i_4 < \cdots < i_{n-2} \leq n$. Let $\tilde{d}_1$ and $\tilde{d}_2$ be the first and second row of D^{-1} respectively and $v^T = [0\ \cdots\ 0\ z\ \ z^2 - z\langle \tilde{d}_1, g_1\rangle\ \ 1 - z^2\langle \tilde{d}_1, g_1\rangle - z\langle \tilde{d}_2, g_1\rangle]$. Then, $A - BD^{-1}C = [\tilde{g}_1\ \cdots\ \tilde{g}_{k-1}\ v]$ and hence, $\det(A - BD^{-1}C) = az^2 + bz + c$ with $c \in \mathbb{F}_q^*$.

Case $(k-2,3,k-1)$: Now, $A = [\tilde{g}_1 \ \cdots \ \tilde{g}_{k-2} \ x_1 \ x_2]$, $B = [x_3 \ 0 \ \cdots \ 0]$, $C = [0 \ \cdots \ 0 \ g_1 \ g_2]$ and $D = [g_3 \ g_{i_4} \ \cdots \ g_{i_{n-2}}]$ with g_{i_j} the i_j-th column of G_0, $4 \leq i_4 < \cdots < i_{n-2} \leq n$. For $v_1^T = [0 \ \cdots \ 0 \ \ z \ \ z^2 \ \ 1 - z\langle\tilde{d}_1, g_1\rangle]$ and $v_2^T = [0 \ \cdots \ 0 \ \ z \ \ z^2 - z\langle\tilde{d}_1, g_2\rangle]$, $A - BD^{-1}C = [\tilde{g}_1 \ \cdots \ \tilde{g}_{k-2} \ \ v_1 \ \ v_2]$ and $\det(A - BD^{-1}C) = a(z^4 - z) + bz^2 + cz^3$ with $a \in \mathbb{F}_q^*$. This is nonzero if and only if $p(z) = z^3 + \frac{c}{a}z^2 + \frac{b}{a}z - 1 \neq 0$, which is true since the minimal polynomial of z has degree 3 and is not equal to $p(z)$. □

Recently, [Che26] provided a construction for MDP codes with $L = 1$ and $\delta = 2, 3$, which requires a smaller field size for $\delta = 2$, whereas our construction requires the smallest field size among the existing constructions for MDP codes with these parameters for $\delta = 3$; see [DLSS26] for details.

Theorem 9. *[DLM+25] The encoding of the MDP codes from Theorem 6, Theorem 7 and Theorem 8 requires $O(n \log^2(n))$ multiplications.*

Proof. For the codes of Theorem 6 this has been shown in [DLM+25] and for the other codes it can be shown in a similar way. □

5 Conclusion

We provided constructions for MR-LRCs with UL and MDP codes with similar building blocks, i.e., MDS matrices over some base field and diagonal/ lower-triangular matrices over an extension field, using similar proof strategies. However, even if the two corresponding matrix completion problems both require to construct a matrix with a given zero pattern such that all non-trivial full-size minors of this matrix are nonzero, the differences in the constraints cause the building blocks to be arranged in a different way. Thus, independent proofs are required and we do not have a direct way of obtaining an MDP code from an MR-LRC with UL or the other way around. Hence, the results of this paper can be seen as a first step towards reaching the goal mentioned in the Introduction, i.e., to develop principled matrix completion frameworks that apply uniformly across a wide range of constraint patterns. As a next step, we aim to develop constructions for MDP codes with $L > 1$ and MR-LRCs with UL without the restriction on the number of global parities.

For future research we also want to investigate to which extent MDP codes can be seen as MR-LRCs with unequal and hierarchical locality [NL19, RB26]. Moreover, in [MPN20] partial MDP codes were considered, where the imposed locality constraints prohibit the codes to be MDP, i.e., the column distances can only reach a modified upper bound adapted to the locality setting, and codes reaching these bounds are called partial MDP. Hence, it is a natural question to investigate whether our techniques can also be used for the construction of partial MDP codes.

References

[ANNR24] G. N. Alfarano, D. Napp, A. Neri, and V. Requena. Weighted Reed–Solomon convolutional codes. *Linear and Multilinear Algebra*, 72(5):841–874, 2024.

[ANP13] P. Almeida, D. Napp, and R. Pinto. A new class of superregular matrices and MDP convolutional codes. *Linear Algebra and its Applications*, 439(7):2145–2157, 2013.

[BHH13] M. Blaum, J. L. Hafner, and S. Hetzler. Partial-MDS codes and their application to RAID type of architectures. *IEEE Trans. Inf. Theory*, 59(7):4510–4519, 2013.

[Che23] Z. Chen. A lower bound on the field size of convolutional codes with a maximum distance profile and an improved construction. *IEEE Transactions on Information Theory*, 70(6):4064–4073, 2023.

[Che26] Z. Cheng. New constructions of MDP convolutional codes with memory 1 for $k = 2, 3$. *Cryptography and Communications*, pages 1–14, 2026.

[CMST22] H. Cai, Y. Miao, M. Schwartz, and X. Tang. A construction of maximally recoverable codes with order-optimal field size. *IEEE Trans. Inf. Theory*, 68(1):204–212, 2022.

[DLM+25] S. Dang, J. Lieb, O. Makkonen, P. Soto, and A. Sprintson. A matrix completion approach for the construction of MDP convolutional codes. In *2025 IEEE Information Theory Workshop (ITW)*, pages 710–715, 2025.

[DLSS26] S. Dang, J. Lieb, P. Soto, and A. Sprintson. Design of MDP convolutional codes and maximally recoverable codes through the lens of matrix completion. *arXiv 2604.21544*, 2026.

[DSY14] S. H. Dau, W. Song, and C. Yuen. On the existence of MDS codes over small fields with constrained generator matrices. In *Proceedings of the IEEE International Symposium on Information Theory (ISIT)*, pages 1787–1791, 2014.

[GG22] S. Gopi and V. Guruswami. Improved maximally recoverable LRCs using skew polynomials. *IEEE Trans. Inf. Theory*, 68(11):7198–7214, 2022.

[GGS87] A. Gerasoulis, M. D. Grigoriadis, and L. Sun. A fast algorithm for Trummer's problem. *SIAM Journal on Scientific and Statistical Computing*, 8(1):s135–s138, 1987.

[GHJY14] P. Gopalan, C. Huang, B. Jenkins, and S. Yekhanin. Explicit maximally recoverable codes with locality. *IEEE Trans. Inf. Theory*, 60(9):5245–5256, 2014.

[GLRS06] H. Gluesing-Luerssen, J. Rosenthal, and R. Smarandache. Strongly MDS convolutional codes. *IEEE Trans. Inf. Theory*, 52(2):584–598, 2006.

[HTN20] A.-L. Horlemann-Trautmann and A. Neri. A complete classification of partial MDS (maximally recoverable) codes with one global parity. *Advances in Mathematics of Communications*, 14(1):69–88, 2020.

[KS16] S. Kadhe and A. Sprintson. Codes with unequal locality. In *2016 IEEE International Symposium on Information Theory (ISIT)*, pages 435–439, 2016.

[LCEL23] G. Luo, X. Cao, M. F. Ezerman, and S. Ling. A construction of maximum distance profile convolutional codes with small alphabet sizes. *IEEE Transactions on Information Theory*, 69(5):2983–2990, 2023.

[Lov21] S. Lovett. Sparse MDS matrices over small fields: A proof of the GM-MDS conjecture. *SIAM Journal on Computing*, 2021.

[MP22] U. Martinez-Penas. A general family of MSRD codes and PMDS codes with smaller field sizes from extended Moore matrices. *SIAM Journal on Discrete Mathematics*, 36(3):1868–1886, 2022.

[MPN20] U. Martínez-Peñas and D. Napp. Locally repairable convolutional codes with sliding window repair. *IEEE Transactions on Information Theory*, 66(8):4935–4947, 2020.

[NL19] A. M. Nair and V. Lalitha. Maximally recoverable codes with hierarchical locality. In *2019 National Conference on Communications (NCC)*, pages 1–6, 2019.

[PKLK12] N. Prakash, G. M. Kamath, V. Lalitha, and P. V. Kumar. Optimal linear codes with a local-error-correction property. In *2012 IEEE International Symposium on Information Theory Proceedings*, pages 2776–2780, 2012.

[RB26] R. P. Rajpurohit and M. Bhaintwal. A new construction of maximally recoverable codes with hierarchical locality. *Finite Fields and Their Applications*, 109:102686, 2026.

[TRS12] V. Tomás, J. Rosenthal, and R. Smarandache. Decoding of convolutional codes over the erasure channel. *IEEE Transactions on Information Theory*, 58(1):90–108, 2012.

[YH19] H. Yildiz and B. Hassibi. Optimum linear codes with support-constrained generator matrices over small fields. *IEEE Transactions on Information Theory*, 2019.

[Yor97] E.V. York. *Algebraic Description and Construction of Error Correcting Codes: A Linear Systems Point of View.* Ph.D. dissertation, University of Notre Dame, 1997.

[ZY16] A. Zeh and E. Yaakobi. Bounds and constructions of codes with multiple localities. In *2016 IEEE International Symposium on Information Theory (ISIT)*, pages 640–644, 2016.